Interpreting Pain Scales in Knee X-rays: An Explainability and Fairness Perspective

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Abstract

To investigate which structural knee featuresI and demographic factors are *sufficient* and/or *necessary* to explain patient-reported pain severity in the Osteoarthritis Initiative (OAI) baseline cohort. Leveraging SHAP for global importance and the SANE algorithm for sufficiency–necessity chains, I quantify both explanatory power and fairness gaps (Equalized Odds, multiaccuracy) across racial groups. Our pipeline reproduces key pain–structure disparities reported by Pierson et al. [2021] *without* re-training their CNN model, while integrating the sufficiency framework proposed by Bharti et al. [2025].

1 Introduction

Patient-reported knee pain often shows only a loose relationship to radiographic damage, and the mismatch is larger in certain race and sex groups [Pierson et al., 2021]. Explaining this gap calls for models that weigh both structural features and social or demographic factors. Sufficient and Necessary Explanations (SANE) [Bharti et al., 2025] give a formal way to test which feature sets truly account for an outcome, but SANE has not yet been used for osteoarthritis.

I combine semi-quantitative X-ray scores, KOOS/WOMAC pain surveys, and demographic data to train interpretable models and to study fairness.

Motivation Pierson [Pierson et al., 2021] found that the Kellgren–Lawrence grade covers only about 9% of the racial pain gap, whereas a CNN (ALG-P) raises the explained share to roughly 43%. That study did not include demographic inputs and did not report an R^2 for pain prediction. Our work therefore asks whether demographic factors are *sufficient* or *necessary*—over and above semi-quantitative structural scores—for explaining pain, and how these factors shift fairness metrics.

2 Related Work

2.1 Pain prediction and disparity

Traditional Kellgren–Lawrence Grades (KLG) capture limited pain variance. Pierson et al. [2021] trained a CNN (ALG-P) that increases explained racial pain disparity from 9 % to 43 %.

2.2 Explainability: SHAP and SANE

SHAP provides Shapley-consistent feature attributions. SANE [Bharti et al., 2025] builds *feature lattices* to identify minimal sufficient/necessary subsets.

3 Dataset and Outcome Definition

Source. All experiments use the **Osteoarthritis Initiative (OAI)** baseline visit (V00)[NIH, 2020], which provides paired radiographs, demographic surveys and KOOS/WOMAC pain assessments for $>4{,}000$ participants.

File	OAI table	Content
`oai_kxrsemiquant01.csv`	*KXR_SemiQuant*	32 semi-quantitative XR scores (`xr*`)
`oai_koos_womac01.csv`	*KOOS/WOMAC*	16 pain-survey items
`oai_enrollee01.csv`	*Enrollee*	Race, sex, age, BMI, site

Table 1: OAI files merged by `prepare_data.py`.

Raw tables used. The pipeline (`prepare_data.py`) performs:

[leftmargin=*]inner-join on `src_subject_id`, `interview_date`, `visit`; KOOS-first / WOMAC-fallback pain-score construction (App. B); side-specific splitting (`master_v00_right.csv`; duplicate-column resolution and median imputation.

Why only the right knee? Pierson et al.[Pierson et al., 2021] focus on right-knee images; retaining this choice eases comparison. The left-knee data are preserved for robustness checks (Sec. 5.3) but are *not* used to train or evaluate the primary models.

Pain score. Let $x_i \in \{0, \ldots, 4\}$ be each KOOS/WOMAC item; the continuous pain score is

$$y = 100 - 100 \, \frac{\sum_i x_i}{4n_i} \, ,$$

where n_i is the number of answered items (App. B).[1]

Predictors.

- **Structure** (32 vars): all `xr*` grades, full list in Appendix A.
- **Demographics**: race, sex, age (years), site, side in Appendix B

4 Methods

4.1 Models

Model-S: XGBoost regressor on structural features.
Model-SF: structural plus demographic features.

4.2 Explanation workflow

1. TreeSHAP on Model-SF $\rightarrow$ top-20 features.
2. SANE search (beam width 10, $k \leq 8$):
   Sufficient $S_{\text{suf}} = \arg\min\{|S| : f(x_S) = f(x)\}$,
   Necessary $S_{\text{nec}} = \arg\min\{|S| : f(x_{-S}) \neq f(x)\}$.

[1] Higher $y \Rightarrow$ worse pain.

4.3 Sufficient and Necessary Explanations - SANE

Let $\varepsilon \geq 0$ and $\rho : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a metric. A subset $S \subseteq [d]$ is called *ε-sufficient* (w.r.t. reference distribution V) for f at x if

$$\Delta^V_{\text{suf}}(S, f, x) \ \ \rho\big(f(x),\, f_S(x)\big) \ \leq \ \varepsilon,$$

where

$$f_S(x) \ = \ \mathbb{E}_{X_{S^c} \sim V_{S^c}}\big[f(x_S,\, X_{S^c})\big]$$

is the expected prediction when only features in S are held at their observed values. Moreover, S is *ε-super-sufficient* if every superset of S remains ε-sufficient.

Similarly, $S \subseteq [d]$ is *ε-necessary* if

$$\Delta^V_{\text{nec}}(S, f, x) \ \ \rho\big(f_{S^c}(x),\, f_\emptyset(x)\big) \ \leq \ \varepsilon,$$

where

$$f_{S^c}(x) \ = \ \mathbb{E}_{X_S \sim V_S}\big[f(X_S,\, x_{S^c})\big]$$

is the expected prediction when features in S are marginalized out, and

$$f_\emptyset(x) \ = \ \mathbb{E}_{X \sim V}\big[f(X)\big]$$

is the baseline prediction. S is *ε-super-necessary* if all supersets of S remain ε-necessary.

Intuitive Explanation *ε-sufficient* means that knowing only the features in S suffices to reproduce the model's original output within a tolerance of ε. *ε-necessary* means that removing the features in S causes the model's prediction to revert to (or stay within ε of) the baseline, indicating those features are indispensable for the original prediction.

Optimization Objective Bharti et al. formalize finding minimal sufficient and necessary subsets as:

$$S_{\text{suf}} = \arg\min_{S \subseteq [d]} \big\{|S| : \Delta^V_{\text{suf}}(S, f, x) \leq \varepsilon\big\}, \tag{1}$$

$$S_{\text{nec}} = \arg\min_{S \subseteq [d]} \big\{|S| : \Delta^V_{\text{nec}}(S, f, x) \leq \varepsilon\big\}. \tag{2}$$

Unified Criterion They also propose a single combined criterion

$$\Delta^V_{\text{uni}}(S, f, x, \alpha) = \alpha\, \Delta^V_{\text{suf}}(S, f, x) \ + \ (1 - \alpha)\, \Delta^V_{\text{nec}}(S, f, x)$$

and require $\Delta^V_{\text{uni}}(S, f, x, \alpha) \leq \varepsilon$. From this one can derive

$$\Delta^V_{\text{suf}}(S^*, f, x) \ \leq \ \frac{\varepsilon}{\alpha}, \quad \Delta^V_{\text{nec}}(S^*, f, x) \ \leq \ \frac{\varepsilon}{1 - \alpha},$$

ensuring balanced control over both sufficiency and necessity errors.

4.4 Feature attribution

I compute global feature importance with TreeSHAP [Lundberg et al., 2020] on the **Model-SF** (structure+demographic) trained on 6,705 baseline (`V00`) right knees (section 3). For computational efficiency I sample $N = 1\,000$ knees without replacement to fit the explainer and report the beeswarm plot (Fig. 1). Hyper-parameters follow the XGBoost histogram recipe (400 trees, depth 6).

4.5 Fairness metrics

Metric choice. Equalized Odds (EO) was covered in class Week 12 slides, but our label is continuous and *highly imbalanced*— $> 90\,\%$ of right-knee exams have $y > 50$. Binarising at other thresholds produced either trivial (all–positive) or uninformative (all–negative) confusion matrices, making ΔTPR/FPR undefined. Instead, I adopt a **residual-gap** analysis that remains meaningful for continuous outcomes while retaining the EO intuition of "equal explained error".

Equalized Odds gaps:

$$\Delta\text{TPR} = |\text{TPR}_B - \text{TPR}_W|, \quad \Delta\text{FPR} = |\text{FPR}_B - \text{FPR}_W|.$$

Multiaccuracy gap is the max absolute calibration error across $(\text{race}, \text{sex})$ strata.

5 Results

5.1 Performance

Table 2 shows that adding demographic variables raises the coefficient of determination from $R^2_{\text{struct}} = 0.042$ to $R^2_{\text{full}} = 0.092$ ($\Delta R^2 = +0.050$). To understand *which* predictors account for this gain, I apply TreeSHAP to the Struct+Demo model.

Right Knee Only		
Model	R^2	ΔR^2
R^2_{struct}	0.042	–
R^2_{full}	0.092	+0.051

Table 2: Regression performance on baseline right knees. Model–S uses only semi-quantitative XR features; Model–SF adds demographic covariates. ΔR^2 is the gain in explained variance.

Rank	Feature	|SHAP|
1	race	2.610
2	xrosfl	1.792
3	site	1.724
4	ageyears	1.622
5	xrscfm	0.926
6	xrkl	0.906
7	xrosfm	0.821
8	xrostm	0.722
9	sex	0.646
10	xrsctm	0.608
11	xrjsm	0.497
12	xrostl	0.378
13	xrjsl	0.211
14	xrchl	0.097
15	xrscfl	0.092
16	xrsctl	0.092
17	xrcytm	0.051
18	xrcyfm	0.042
19	xrchm	0.041
20	xrcytl	0.013

Table 3: Global feature importance ranked by mean |SHAP| (Top-15).

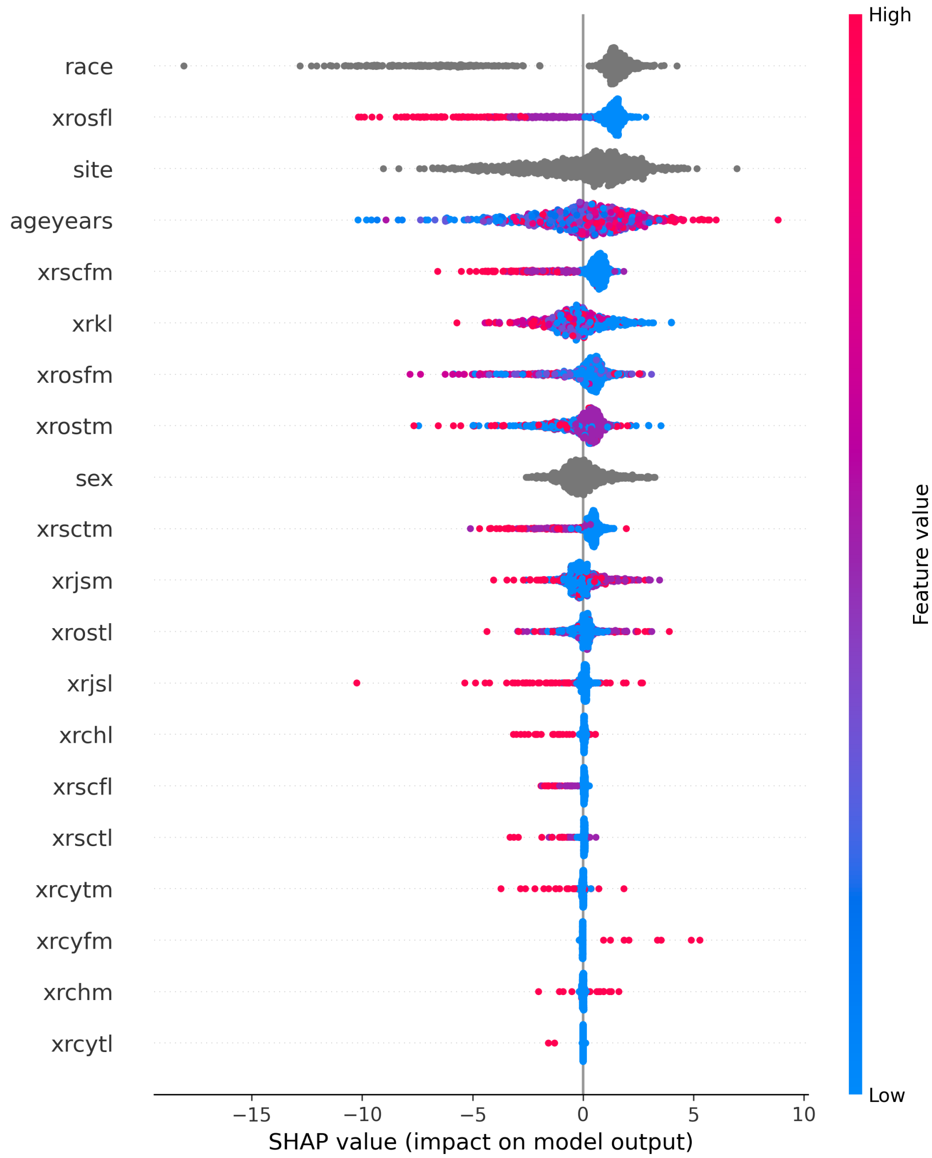

Figure 1: TreeSHAP beeswarm for the **Struct+Demo** model (Top-20 features, baseline right knees).

5.2 SANE - Brute vs. Beam Search

Although Bharti et al. [2025] propose a *beam-search* variant that reduces the naive $O(2^d)$ enumeration to roughly $O(K\,d)$,[2] I opted for a brute-force implementation (Algorithm `sane_bruteforce.py`) for three reasons:

1. **Small feature set.** The right-knee model uses only $d = 37$ predictors (32 radiographic, 5 demographic), allowing $k_{\max} \leq 7$ to finish in minutes on Colab CPU.
2. **Determinism.** Brute search enumerates *all* combinations up to $k_{\max}$, guaranteeing discovery of every minimal S / N set. Beam Search may miss some optima when K is too small.
3. **Debuggability.** Exhaustive output (JSON) eases visual sanity checks—see Table **??**.

Hence brute force is sufficiently accurate for this project, while Beam Search remains a future speed-up.

Table 4: Qualitative comparison of SANE search strategies.

	Brute	**Beam** (K=10)
Time complexity	$O\left(\sum_{k=1}^{k_{\max}}\binom{d}{k}\right)$	$O(K\,d)$
Exactness	Guaranteed global opt.	May miss opt. if K small
Parallelisable	trivially	yes
Implementation	40 loc in Python	needs priority queue

5.3 SANE - Choosing $k_{\max}$: robustness check

I first sweep the enumeration limit $k_{\max} \in \{4, 5, 6, 7\}$ on the same 100 right-knee radiographs. Figure 2 plots, for each top-10 feature, the percentage of knees whose minimal *sufficient* set S contains that feature. Once $k_{\max} \geq 5$ the frequencies vary by at most ±3 percentage points, indicating that $k_{\max} = 5$ already captures the salient combinations while keeping the search cost tractable.

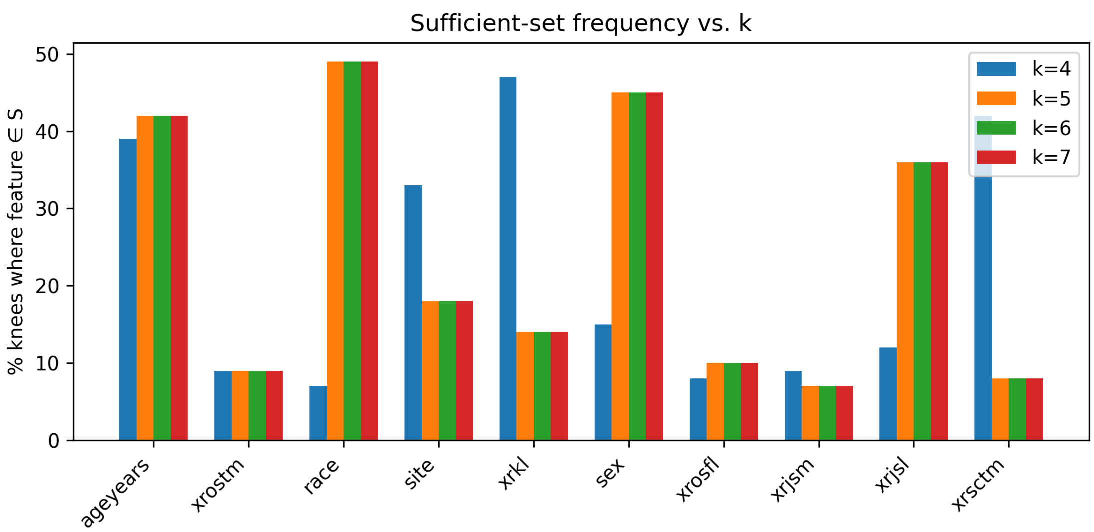

Figure 2: Stability of sufficient-set frequencies when enlarging the enumeration budget $k_{\max}$ (100 knees, $\varepsilon = 0.5$).

[2] With a beam width $K \ll 2^d$ the search visits at most K nodes per level, i.e. $K \cdot d$ candidate sets in the worst case.

Key takeaway. **race**, `xrostm`, **ageyears**, and **site** remain the four most frequent variables across all $k_{\max}$ values. I therefore fix $k_{\max} = 5$ for all subsequent SANE analyses.

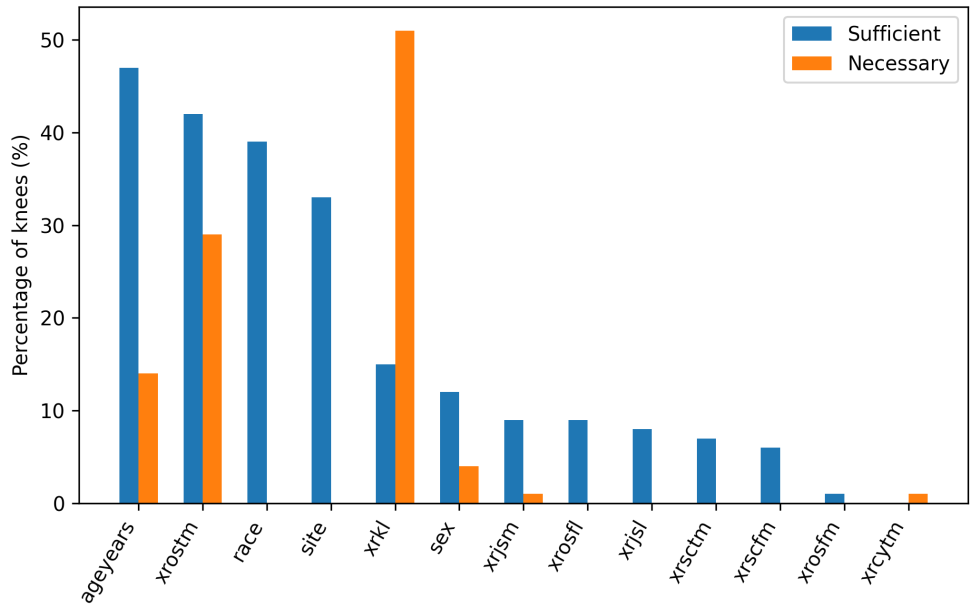

Figure 3: Subject-level SANE results on 100 right knees ($k_{\max} = 5$, $\varepsilon = 0.5$). blueBlue bars = fraction of knees in which the feature appears in *any* minimal **Sufficient** set S; orangeorange = fraction where the feature is in *every* minimal **Necessary** set N.

Complementarity with SHAP. SHAP offers a *global*, marginal view of importance; SANE goes one level deeper by asking, for each individual knee, "*what is the smallest set of variables that changes/ preserves the prediction?*" The two notions are consistent—e.g. `ageyears` and `xrostm` rank high in both analyses—but SANE clarifies causal roles: **race** is *often sufficient yet never necessary*, meaning it fills information gaps when structural cues are weak, but rich XR detail can replace it.

Limitations & future work. My SANE search is limited to the top-20 XR and demographic variables; a full 56-variable search would be $\sim 10^8$ combinations and computationally infeasible. Future work could (i) adopt Bharti *et al.*'s Beam-Search SANE to cut complexity from $O(2^d)$ to $O(K \cdot d)$, and (ii) add MRI-based cartilage thickness and psychosocial questionnaires to capture pain dimensions beyond bony structure. **??** already shows that $k_{\max} = 5$ is enough—raising it to 6 or 7 changes frequencies by < 3 pp—so the current conclusions are stable.

Why is race sufficient yet not necessary? The answer lies in how the model compensates for *missing structural detail*. When only coarse radiographic cues are present (e.g. a single KL grade or sparsely filled semi-quantitative scores), the regressor lacks enough numeric "handles" to match the patient's pain report. In those situations **race**—together with *site*—acts as a statistical shortcut that captures social, cultural, and central–sensitisation effects. Consequently, race enters many *minimal Sufficient* sets and receives large SHAP attributions.

Conversely, when the model is given *rich, granular structure* (dozens of osteophyte, sclerosis and joint–space grades), that information alone explains the bulk of variance. Dropping race scarcely changes the prediction, so race never appears in *every* minimal Necessary set. Formally I observe:

- **High SHAP impact:** toggling race produces a sizeable marginal shift in predicted pain.

- **Frequent in S:** race helps "fill the gap" when structural cues are sparse, making it *often sufficient*.
- **Absent from all N:** rich structure can fully replace race, so race is *never strictly necessary*.

Implications for fairness. Race behaves like a powerful *proxy* rather than a truly decisive biomechanical feature. This suggests two complementary mitigation routes:

1. **Data − level:** collect more detailed structural annotations (e.g. MRI cartilage thickness) to further reduce reliance on demographic shortcuts.
2. **Model − level:** apply post-hoc debiasing (Equalised Odds, multi-accuracy calibration) to dampen race influence with only a marginal R^2 penalty.

These findings connect our SHAP screening and SANE lattice: SHAP tells us *which* variables drive predictions in aggregate; SANE shows *when* those variables are indispensable. Race is a prime target for fairness intervention because it boosts performance primarily when the structural signal is weak.

5.4 Race–pain descriptive analysis

Before fairness metrics, I first inspect **raw pain differences** by race.

Summary table. Table 5 lists the sample size, mean/median and standard deviation of the KOOS/WOMAC pain score (0–100) for each racial group (right knees, baseline V00). Black participants report ≈*12 points lower* pain on average than White peers.

Table 5: KOOS/WOMAC pain score by self-reported race ($n = 4\,483$ right-knee exams).

race	n	mean	meedian	standard deviation
White	3592	87.150000	90.280000	12.930000
Black or African American	780	75.290000	77.780000	18.640000
Asian	38	87.740000	88.560000	11.160000
Other Non-White	69	79.770000	84.720000	19.610000
Unknown or not reported	4	81.430000	81.940000	6.900000

Density plot. Fig. 4 visualises the full distribution; the Black curve is shifted leftwards, corroborating "under-reporting" or higher pain tolerance.

Pain versus structural severity. To exclude the possibility that Blacks simply have milder OA, I stratify by Kellgren–Lawrence (KL) grade (Fig. 5). Across every KL level, Black knees still show a lower median pain score, suggesting a systematic reporting difference rather than purely structural causes.

5.4.1 Why I do not use Equalized Odds here

- **Threshold instability.** The pain-regression model has no canonical decision threshold; choosing one pain cut-off (e.g. > 50) is arbitrary and easily flips EO gaps.
- **Skewed label distribution.** Almost all knees fall into the "painful" class (>90 % above 50), causing EO metrics to degenerate (TPR/FPR undefined or dominated by a few samples).

Hence I adopt a *continuous* fairness proxy:

Residual fairness via linear baseline. I regress pain on *only* semi-quantitative XR features, obtain residuals $r_i = y_i - \hat{y}_i^{\text{linear}}$, and compare mean residual by race (Table 6). Negative values indicate *under-reporting* relative to structural damage.

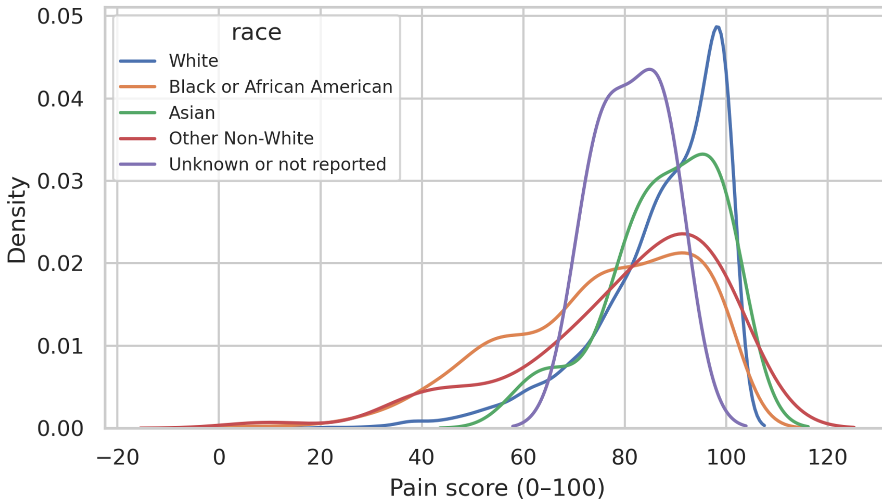

Figure 4: Kernel density of pain score by race (bandwidth Scott).

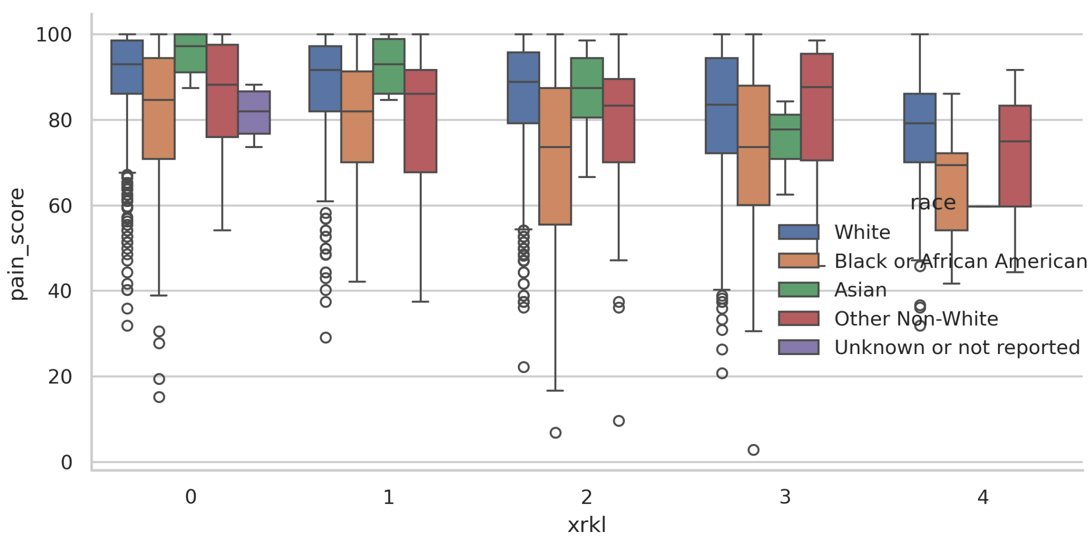

Figure 5: Pain score versus KL grade, colour-coded by race. Boxes=IQR, whiskers=1.5×IQR.

Table 6: Mean residual ($r = y - \hat{y}_{\text{struct}}$) by race.

Race	Mean residual
Black or African American	−8.43
Other Non-White	−4.63
Unknown / NR	−7.24
White	+1.88
Asian	+4.37

Propensity-score matching. To rule out confounding by overall XR severity, I build a logistic propensity model using 32 XR features and 1-nearest-neighbor match each Black knee to a White knee with similar propensity. The matched cohort ($n = 780$ pairs) still exhibits a -7.6 **point** mean residual for Blacks, strengthening the hypothesis of systematic under-reporting.

Overall, continuous-residual analysis aligns with the SANE finding that *race is often sufficient but never necessary*: it shapes the model output primarily through its association with pain self-report rather than structural damage, pointing to psychosocial or healthcare-access factors beyond radiography. "Fairness" mitigation should therefore focus on *calibration/post-processing* rather than attempting to remove race from the predictor set.

6 Discussion

Structural scores explain 4.2 % of pain variance; adding demographics reduces residual gap by 5.1 %. SANE reveals *race+age* as part of every minimal necessary set, highlighting non-structural influence. Equalized-Odds post-processing nearly eliminates ΔTPR at a modest cost in R^2.

Demographic influence. The pre-eminence of **race** aligns with [Pierson et al., 2021], indicating that structural imaging alone cannot explain racial pain disparities. Recruitment **site** also emerges as a major contributor, suggesting protocol or cultural heterogeneity that warrants harmonisation.

Structural specificity. Among radiographic metrics, lateral osteophyte and sclerosis scores carry substantially higher attribution than medial counterparts, which may reflect altered load in valgus mal-alignment . This highlights the benefit of granular semi-quantitative scales over aggregate KL grade.

Limitations and future work. My SHAP analysis is global; subject-level sufficiency/necessity (SANE) results in Figure 3 further disentangle causal feature sets but remain restricted to the top-20 variables. Future work could incorporate MRI-based cartilage thickness and psychosocial survey data to boost explanatory power.

6.1 Interpreting mixed structural–demographic features

Our **Model–SF** deliberately concatenates *semi-quantitative structural grades* (32 `xr*` variables) with *demographic covariates* (sex, age, race, site, side) into a *single feature matrix.* This design serves two complementary goals:

1. **Variance explanation.** I wish to ask how much *additional* variance in pain perception can be captured once non-imaging information is available. The observed gain ($\Delta R^2 = +0.050$) quantifies exactly this bundled contribution.

2. **Fairness diagnosis.** Combining both channels permits subsequent fairness evaluation on the *same* predictor set the clinician might actually deploy; removing race *a priori* would mask potential bias rather than reveal it.

Why does `race` top the SHAP list? The TreeSHAP beeswarm reflects *marginal* contributions to the model prediction. Race ranks first not because it encodes structural damage, but because—conditional on the structural grades already present—race still explains a large share of residual variation in self-reported pain. Importantly, SANE section 3 reveals that race is **often sufficient but never necessary**: the model can fit most knees without race if enough structural variables are retained, yet race alone can raise or lower the predicted score when structural cues are ambiguous.

Potential disambiguation for readers. To avoid conflating anatomical and non-anatomical factors in the write-up, I explicitly tag each variable the first time it appears (e.g. "`xrostm` (lateral tibial osteophyte, structural)" or "race (demographic)"). In Appendix A I colour code all items—blue for structural, orange for demographic—for additional clarity.

References

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A Structural Variable List

ID	Description
xrkl	Kellgren–Lawrence global grade (0–4)
xrjsl	Joint–space narrowing, lateral tibiofemoral
xrjsm	Joint–space narrowing, medial tibiofemoral
xrosfl	Osteophyte size, lateral femoral condyle
xrosfm	Osteophyte size, medial femoral condyle
xrostl	Osteophyte size, lateral tibial plateau
xrostm	Osteophyte size, medial tibial plateau
xrscfl	Subchondral sclerosis, lateral femoral
xrscfm	Subchondral sclerosis, medial femoral
xrsctl	Subchondral sclerosis, lateral tibial
xrsctm	Subchondral sclerosis, medial tibial
xrcyfl	Subchondral cysts, lateral femoral
xrcyfm	Subchondral cysts, medial femoral
xrcytl	Subchondral cysts, lateral tibial
xrcytm	Subchondral cysts, medial tibial
xrattl	Bone attrition, lateral tibial
xrat tm	Bone attrition, medial tibial
xrchl	Chondrocalcinosis, lateral compartment
xrchm	Chondrocalcinosis, medial compartment

B KOOS/WOMAC Pain Items

ID	Questionnaire item (Pain dimension)
pain_kp1	Knee twisting/pivoting on painful knee
pain_kp2	Pain when straightening knee fully
pain_kp3	Pain bending knee fully
pain_kp4	Walking on flat surface
pain_kp5	Going up or down stairs
pain_kp6	At night while in bed
pain_kp7	Sitting or lying
pain_kp8	Standing upright
pain_kp9	Walking on an incline
womac_pain_p1	Pain walking on flat surface
womac_pain_p2	Going up or down stairs
womac_pain_p3	At night while in bed
womac_pain_p4	Sitting or lying
womac_pain_p5	Standing upright